Annals of Botany **83**: 645–654, 1999
Article No. anbo.1999.0866, available online at http://www.idealibrary.com on IDEAL®

Green Fluorescent Protein Targeted to the Nucleus, a Transgenic Phenotype Useful for Studies in Plant Biology

EVA CHYTILOVA*†, JIRI MACAS*‡ and DAVID W. GALBRAITH§

* *Department of Plant Sciences, University of Arizona, 303 Forbes Building, Tucson, Arizona 85721, USA,*
† *Masaryk University, Department of Genetics and Molecular Biology, Brno, Czech Republic and*
‡ *Institute of Plant Molecular Biology, Ceske Budejovice, Czech Republic*

Received: 21 August 1998 Returned for revision: 29 October 1998 Accepted: 17 February 1999

We present a characterization of transgenic *Arabidopsis thaliana* (L.) Heynh. plants expressing a chimeric gene comprising the Green Fluorescent Protein (GFP) and β-glucuronidase (GUS) coding sequences, fused to an efficient nuclear localization signal (NLS). The transgenic plants accumulate the fusion protein in their nuclei, and this provides a novel phenotype, that of green-fluorescent nuclei. The fluorescent nuclei are readily observed using conventional epifluorescence and laser scanning confocal microscopy. We describe the use of this phenotype for *in vivo* studies of nuclear shape and movement, cell division, and for analysis of the transcriptional activities of constitutive and tissue-specific promoters. We propose that the phenotype of fluorescent nuclei will prove particularly valuable in histological, physiological and developmental studies of higher plants that require the facile observation of nuclei within living cells and in the absence of fixation or external staining.

© 1999 Annals of Botany Company

Key words: *Arabidopsis thaliana*, GFP, nucleus, D35S promoter, CoYMV promoter, KAT1 promoter.

INTRODUCTION

One of the current goals of biological research is to develop an understanding of the mechanisms connecting the information contained in the genome, the three-dimensional structure of cells and tissues resulting from expression of this information, and the way in which this information changes over time. An important step towards achieving this goal would be to devise methods which simply and accurately report the presence, amounts and three-dimensional locations of all of the different protein components of living cells during growth and development. Such an ambitious goal is not readily achieved. However, many of the proteins in eukaryotic cells comprise parts of supra-molecular structures, termed organelles. It is clear that changes in organelle structure occur during normal growth and development, and in response to external perturbation. Correspondingly, the ability to chart, *in vivo*, the morphological and positional responses of organelles to developmental and environmental cues would be a first step towards providing information about these processes. The work described here provides this ability.

For some classes of organelles, changes can be visualized based on intrinsic optical properties. Thus, chloroplasts are pigmented and exhibit an intense, chlorophyll-based auto-fluorescence, and their positions, numbers, sizes, protein composition, and development from pro-plastids can be readily monitored using techniques based on conventional light and fluorescence microscopy and flow cytometry (see, for example, Galbraith, Harkins and Jefferson, 1988; Pyke and Leech, 1994; Galbraith and Lambert, 1996; Leech and Marrison, 1996). Most of the remaining organelles in the cell are non-pigmented, and are difficult or impossible to monitor *in vivo* unless highlighted using specific fluorescent dyes, for example mitochondria using rhodamine 123 (Wu, 1987), and the endoplasmic reticulum using 3,3-dihexyloxacarbocyanine iodide (Knebel, Quader and Schnepf, 1990). There are no methods at present to monitor the position *in vivo* of the nucleus using fluorescent microscopy. In the first case, nuclei are translucent and non-fluorescent; in the second, fluorescent dyes that specifically stain nuclei do not readily penetrate into living plant cells, particularly when organized as three-dimensional tissues. It should be noted that although nuclei are recognizable using certain forms of bright-field microscopy (for example under Differential Interference Contrast (Nomarski) conditions), this form of microscopy is seriously compromised when applied to organized plant tissues, a consequence either of the high intrinsic refractive index of some tissue components, or of the presence of large numbers of pigmented organelles (such as chloroplasts). This is a major deficiency in our scientific methods, and systematic studies concerning the positional and dynamic behaviour of plant nuclei have been restricted to translucent tissues (see, for example, Inoué and Oldenbourg, 1998 and references therein).

The Green Fluorescent Protein (GFP) of the jellyfish *Aequorea victoria* has recently emerged as a novel genetic marker than can be directly visualized in living cells of many different organisms (Haseloff and Amos, 1995; Tsien, 1998; Sullivan and Kay, 1999). GFP has many characteristics which make it a particularly convenient marker. Formation

§ For correspondence. Fax (520) 621-7186, e-mail galbraith@arizona.edu

0305-7364/99/060645+10 $30.00/0

© 1999 Annals of Botany Company

of the fluorescent chromophore occurs as an intramolecular reaction sequence that is limited only by the availability of molecular oxygen; it is independent of cellular co-factors (Chalfie *et al*., 1994; Haseloff and Amos, 1995). As a general rule, transgenic expression of GFP within any given cell requires simply placing the GFP coding sequence (or slightly modified versions of this sequence) under the transcriptional control of appropriate regulatory sequences (Tsien, 1998; Sullivan and Kay, 1999). GFP is photostable, tolerates fusions with other proteins (Wang and Hazelrigg, 1994; Tsien, 1998), and can be readily detected under the microscope and using other methods of quantitative fluorescence analysis, such as flow cytometry (Tsien, 1998; Sullivan and Kay, 1999). In higher plants, most reports employ GFP variants in which the codon usage is altered to be closer to the plant consensus, which has the additional effect of eliminating aberrant mRNA processing due to false recognition of cryptic introns by some plant species (Haseloff *et al*., 1997). Specific replacements of individual amino acids in the GFP sequence can improve the fluorescent yield and photostability of the fluorescence, and provide variants with altered spectral properties (Tsien, 1998; Haseloff, 1999). GFP has been widely used as a transcriptional reporter and cell marker in many different organisms, and as a fusion partner to monitor the localization and fate of different proteins (Tsien, 1998; Sullivan and Kay, 1999). GFP has also been employed as an active indicator for examination of the distances between proteins or polypeptide domains using fluorescence resonance energy transfer (Tsien, 1998). GFP expression has found a variety of applications in higher plants, first as a general transgenic marker (Galbraith *et al*., 1995; Sheen *et al*., 1995; Chiu *et al*., 1996; Pang *et al*., 1996; Reichel *et al*., 1996; Haseloff *et al*., 1997; Rouwendal *et al*., 1997), and for other purposes including studies of virus invasion and spread (Baulcombe, Chapman and SantaCruz, 1995; Oparka *et al*., 1997), visualization of targeting of cellular proteins (Grebenok, Lambert and Galbraith, 1997*a*; Grebenok *et al*., 1997*b*; Haseloff *et al*., 1997; Köhler *et al*., 1997 *a*, *b*), characterization of cell type-specific gene regulation (Haseloff, 1999) and monitoring of transgenic plants in crops (Stewart, 1996).

In this paper we describe and analyse a novel phenotype in *Arabidopsis thaliana* that allows positional monitoring of the nucleus in living tissues. It results from transgenic expression of GFP fused to a plant nuclear localization signal (NLS) and to β-glucuronidase (GUS) from *E. coli*. Fusion to the GUS coding sequence increases the size of the chimeric protein such that it is prevented from passive movement across the nuclear pores (Grebenok *et al*., 1997*a*, *b*). Accumulation of NLS-GFP-GUS can be easily monitored using fluorescence and confocal microscopy, and transgenic expression is non-toxic. Some specific examples are described that highlight the value of this fluorescent phenotype.

MATERIALS AND METHODS

Construction of plasmids

As a recipient for the various plant promoters, a binary vector plasmid pBGF-0 containing the promoterless NLS-sGFP-GUS coding sequence was constructed from plasmids pRJG23 (Grebenok *et al*., 1997*b*) and pBin19 (Bevan, 1984). The unique *Eco*RI site of pRJG23 was first eliminated by cutting, filling in and re-ligation. The NLS-sGFP-GUS fragment excised by *Hin*dIII and *Xho*I was ligated to a linker containing *Hin*dIII and *Bam*HI termini and an internal *Xba*I site. The chimeric fragment was then ligated with *Sal*I/*Bam*HI-cut pBin19 to give pBGF-0. This plasmid contains several unique restriction sites (*Eco*RI, *Sac*I, *Sma*I, *Bam*HI, *Xba*I), which are followed by the NLS-sGFP-GUS-polyA sequence of pRJG23.

The following plasmids were constructed by ligating specific promoter sequences into the unique sites of pBGF-0: (1) pBGF-KAT contains a 3·4 Kb fragment of the KAT1 promoter (Nakamura *et al*., 1995). This fragment was obtained by PCR amplification of the corresponding sequence from the pKATGA4/KATG8-1 plasmid (obtained from R. E. Hirsch), using the primers Kat1-F (5′ CAC GAG CTC TCT CAT ATA AAT CAT GCC GAC ATT AC 3′) and Kat1-R (5′ TTT CGG GAT CCA GAG ATC GAC AGC TTT TTG A 3′), and was ligated into *Sst*I/*Bam*HI-cut pBGF-0; (2) pBGF-CoY contains the 1040 bp CoYMV promoter (Medberry, Lockhart and Olszewski, 1992) amplified from pColbam (obtained from N. E. Olszewski) using the primers CoYMV-F (5′ CAT CGA ATT CTT AGG GGC TTC TCT C 3′) and CoYMV-R (5′ CAT CGG ATC CGT TGT TGT GTT GGT TTT CTA AGC 3′), and ligated into *Eco*RI/*Bam*HI-cut pBGF-0; (3) pBGF-D35S contains the doubled CaMV 35S promoter, excised from pRJG23 using *Sst*I/*Hin*dIII, and ligated into *Sst*I/*Xba*I-cut pBGF-0 after filling in the *Hin*dIII and *Xba*I sites.

Plant transformation

Agrobacterium tumefaciens strain EHA105 was transformed by electroporation, using kanamycin (60 mg l^{-1}) for selection. Transformation of *Arabidopsis thaliana* (ecotypes Wassilewskaja and Columbia) was done by vacuum infiltration (Bechtold, Ellis and Pelletier, 1993; Bent *et al*., 1994). Seedlings resistant to kanamycin (50 mg l^{-1}) were screened for GFP expression using confocal microscopy. Seedlings expressing GFP were transferred to the greenhouse, and the seeds were collected from self-pollinated plants.

Microscopic observations

Confocal microscopy was done using a BioRad 1024 confocal scanning head attached to a Nikon Optiphot 2 microscope equipped with PlanApo objectives (4×, 10×, 20× and 40×), as previously described (Grebenok *et al*., 1997*a*, *b*). Excised organs were scanned as whole tissue mounts in water on standard slides covered by 22 × 50 mm micro cover glasses (VWR Scientific, PA). Time-lapse images were obtained following germination of transgenic seeds on MS medium (Murashige and Skoog, 1962) containing 0·5% sucrose solidified with 1·5% Phytagar (Gibco BRL, NY) in Lab-Tek Chamber Slide System–1 Well Chambered Coverglass (Nunc, Nalgene, NY). In this arrangement, the primary

root grew downwards until it encountered the coverslip comprising the lower surface of the chamber. The roots were then observed by confocal scanning through the coverslip. Time-lapse studies were employed in the analysis of cell division and of nuclear movement. TIFF images were exported using Lasersharp software (BioRad, Hercules, CA), processed using Adobe Photoshop, and recorded using a Tektronix Phaser 440 dye sublimation printer.

RESULTS

In this study we have employed nuclear-targeted GFP as a means of examining the shape, position, and dynamic behaviour of nuclei in various *Arabidopsis* tissues and cell types. Analysis of GFP fluorescence is done *in vivo*, and visualization of the nuclei does not require staining. This eliminates potential artefacts associated with the use of external dyes and tissue fixation. GFP was expressed in plants under the transcriptional control of two classes of promoter. Use of the constitutive CaMV 35S promoter (Benfey and Chua, 1990) provided labelling of nuclei in the majority of the tissues of the plant, and allowed a general description of nuclear size and shape, position, and movement within these tissues. Use of tissue-specific promoters allowed examination of nuclei within their corresponding cell types, and provided information about the regulation of these promoters during development of these specific cells. The transgenic plants that were produced were normal in appearance, development and fertility. The strategy of highlighting plant nuclei with GFP is formally equivalent to that recently described for plant mitochondria and chloroplasts using appropriate targeting sequences (Köhler *et al*., 1997*a*, *b*).

Constitutive expression of nuclear-targeted GFP

A total of 15 kanamycin-resistant transgenic plant lines were produced following transformation with pBGF-D35S. Seven of these expressed high levels of GFP fluorescence, which was exclusively localized in the nuclei, and could be readily observed by stereomicroscopy under conditions of epifluorescence illumination. Confocal examination of these transgenic plants indicated the CaMV 35S promoter was active in most tissue types (Fig. 1A, B, D–F). The expression levels in old leaves were much lower than in young ones; this phenomenon was particularly evident at flowering.

Nuclear shape, size and movement. Nuclei within plants differed considerably in size and shape. The shape ranged from spherical (Fig. 1A and B), the most commonly observed nuclear morphology, to a highly elongated, almost rod-like shape, for the nuclei of the vascular tissues (Fig.

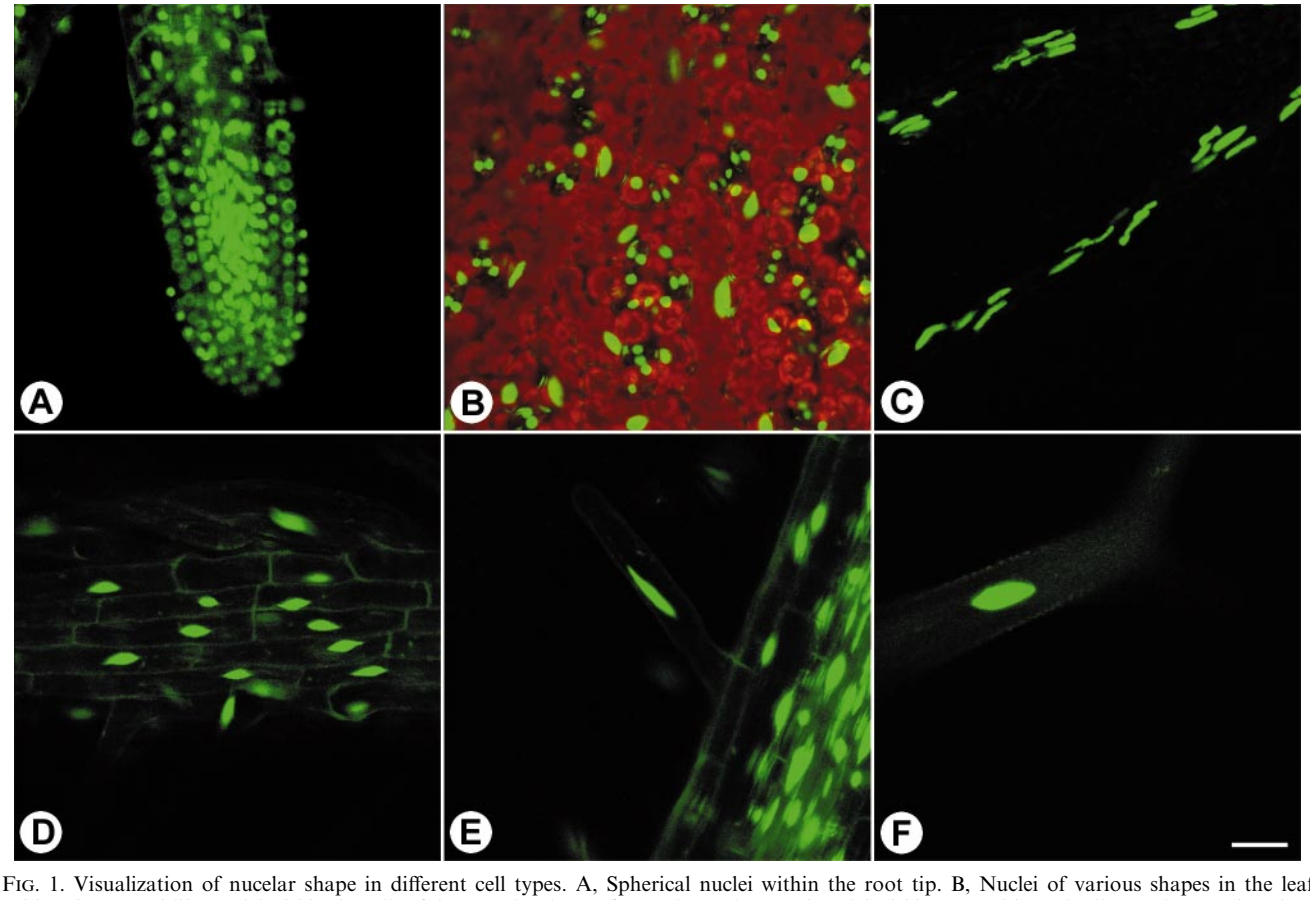


FIG. 1. Visualization of nucelar shape in different cell types. A, Spherical nuclei within the root tip. B, Nuclei of various shapes in the leaf epidermis. C, Rod-like nuclei within the cells of the vascular tissue of a petal. D, Elongated nuclei within root epidermal cells. E, Elongated nucleus within a root hair. F, Oval nucleus within a leaf trichome. All plants were transformed using the pBGF-D35S vector, with the exception of the plant in C, which was transformed using the pBGF-CoY vector. Bar = 25 μm.

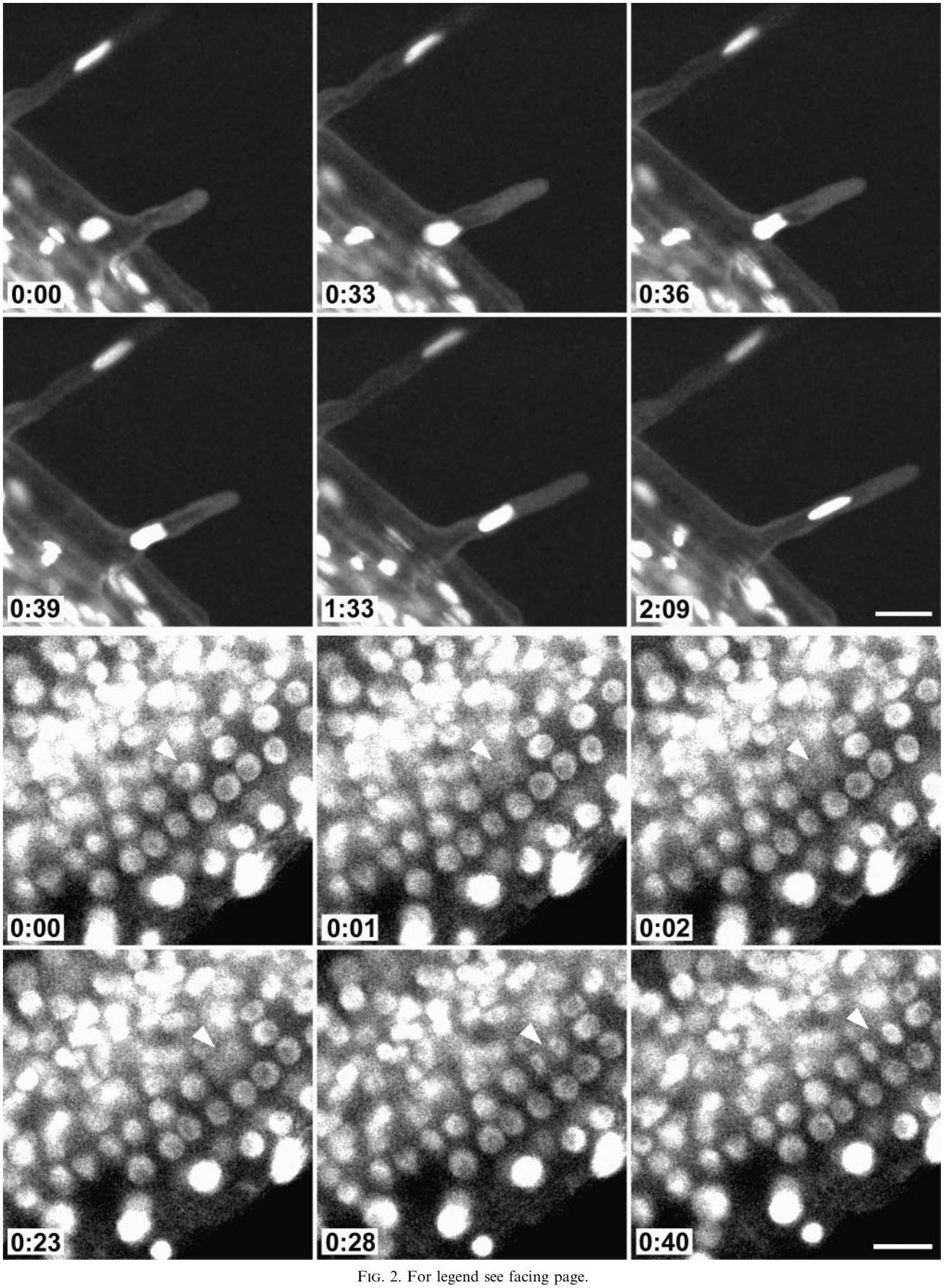


FIG. 2. For legend see facing page.

1C). The elongated shape of these nuclei was similar to the shape of their cells, a situation that was also observed in other anisometric cell types such as root epidermis cells (Fig. 1D) and root hairs (Fig. 1E).

Variation in nuclear size was also observed in *Arabidopsis* plants. These differences were most apparent in the leaf epidermis (Fig. 1B), where the nuclei of guard cells were clearly the smallest, and were almost completely surrounded by cells having bigger nuclei, oval in shape. The largest nuclei were those found within trichomes (Fig. 1F). They appeared much larger than the rest of the nuclei within the plants, and were oval in shape.

The presence of the GFP fusion protein exclusively within the nuclei enabled us to observe the patterns of intracellular movement of nuclei in different tissues *in vivo*. In most tissues, this movement did not show obvious patterns. An exception was found in the root hairs, where the movement of nuclei was under obvious temporal regulation. Using time-lapse microscopy it was possible to monitor the position of the nucleus during the process of root hair formation (Fig. 2A–F). Initially, the nucleus was found to reside at a medial position within the epidermal cell. During expansion of the apical swelling, the nucleus moved from its medial epidermal position to the root hair base, until it reached a point directly below the root hair. It then rapidly migrated into the root hair body. In the process of entering into the root hair, the nucleus lost its spherical appearance, becoming thinner and more elongated. As the root hair continued elongation, the nucleus moved to the middle part of the hair and elongated further (data not shown).

The process of cell division. Nuclear targeting of GFP permits analysis of the events accompanying mitosis, karyokinesis and cytokinesis. We examined the tips of primary roots using time-lapse confocal microscopy. We were able to observe cortical cells close to the root tip undergoing cell division. The whole process, from the disappearance of the G2 nucleus to the re-emergence of the two G1 daughter nuclei, took approx. 30 min (Fig. 2G–L). The act of nuclear envelope disassembly, which defines entry into prometaphase, was very rapid. Between adjacent frames (corresponding to a 1 min time interval), the discrete fluorescent nucleus in the mother cell disappeared, being replaced by a dim green cloud which spread rapidly throughout the volume of the cell (Fig. 2H). The situation remained unchanged for the next 20 to 30 min. Subsequently, the two nuclei of the daughter cells became visible (Fig. 2K). The intensity of intranuclear fluorescence increased over time, and was accompanied by physical separation of the nuclei.

Cell type-specific expression

In further experiments, we produced transgenic *Arabidopsis* plants in which expression of the NLS-GFP-GUS was regulated by two promoters whose activity has previously been described as cell type-specific. These were the *Commelina* yellow mottle virus (CoYMV) promoter (Medberry *et al*., 1992), and the *Arabidopsis thaliana* KAT1 promoter (Nakamura *et al*., 1995).

Expression regulated by the CoYMV promoter. GFP expression was observed predominantly in vascular tissues, and was under developmental regulation, with the highest levels being found in young plants. For most of the transformed lines, expression in 1-week-old plants was predominantly located within the vascular tissues of roots (Fig. 3A and B), stems and leaves (Fig. 3C). All lines exhibited expression in some of the mesophyll cells, as well as in various other tissue types (individual lines typically showed expression in some cells of one or more of the following tissue types: root hairs, the epidermis of the root tip, and the guard cells of the cotyledons and stems). In one of the 15 lines examined, expression in the leaf trichomes accompanied expression in the vascular tissues; however, no expression was seen in the trichomes of the stem or sepals. GFP was no longer detectable in the guard cells of fully-developed rosette leaves and was restricted to leaf veins, being uniformly distributed along the length of the leaves. Upon transition from vegetative growth to flowering, the level of expression in leaves decreased, resulting in a gradient of GFP expression, with the highest expression occurring at the leaf base, and the lowest at the leaf apex. During flowering, GFP expression was initially concentrated in the flower buds and the adjacent small leaves. At the earliest time points, expression was restricted to the base of the buds. As they expanded, GFP expression spread throughout the buds (Fig. 3D). At flower opening, no evidence remained of GFP expression in the leaves. However, within the flowers, all parts expressed GFP, with this expression being restricted to the vascular tissue. In anthers, expression was localized to the vascular bundle within the region separating the anther lobes (Fig. 3E). It was not possible to define patterns of GFP expression within the anthers more precisely, due to the presence of a low level of (non-GFP) autofluorescence in these tissues.

Five out of 15 lines had no expression in the root vascular tissues, but expressed GFP in non-vascular tissues at the root tip, and in some of the root hairs located at the root/shoot interface. These lines displayed generally weak expression of GFP in the whole plants.

Expression regulated by the KAT1 promoter. Transgenic expression of GFP under the control of the KAT1 promoter was detected only in the guard cells of stomata. We did not observe GFP expression in any other cell types. In general, the intensity of KAT1-regulated expression was much weaker than that found for the D35S and CoYMV promoters. Expression was observed in all tissues containing stomata, including cotyledons, leaves (Fig. 3F), sepals and pistils. An interesting feature of expression in pistils was

FIG. 2. A–F, Time-lapse analysis of a longitudinal optical section of a transgenic root, demonstrating the movement of the nucleus during root hair development. Images were collected at the indicated time points (h:min). Bar = 25 μm. G–L, Time-lapse analysis of a longitudinal optical section of a transgenic root tip, illustrating the process of cell division. Images were collected at the indicated time points (h:min). The arrows indicate the dividing nucleus of the mother cell, and, at the later time point, the corresponding nuclei of the two daughter cells. Bar = 10 μm. In both cases, the plants were transformed using the D35S-GFP-GUS vector.

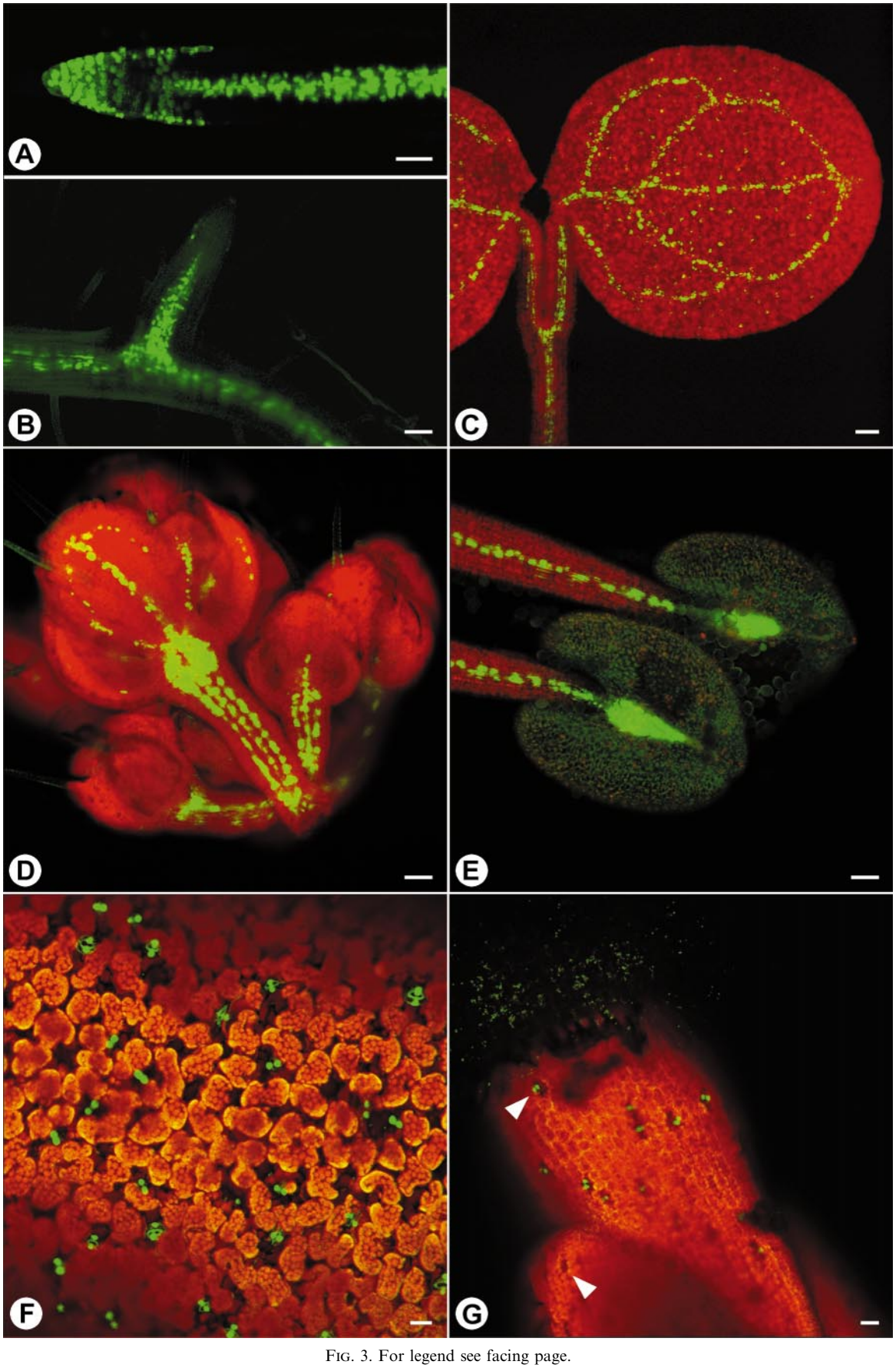


FIG. 3. For legend see facing page.

that GFP was concentrated in the style (Fig. 3G). In lines having the highest levels of GFP expression, it was possible to detect GFP in guard cells located on the sides of carpels close to the styles, but expression in these cells was very weak, and was limited to a subset of these guard cells. We detected no GFP expression in guard cells on carpels in lines having a moderate or low general level of GFP expression. This pattern of expression was consistently observed in different transgenic plants, therefore appearing a typical feature of expression regulated by the KAT1 promoter. One transgenic plant displayed an unusual and interesting phenotype: GFP expression was not seen in all guard cells and, furthermore, was often observed in only one of the two guard cells forming the individual stomatal complexes (data not shown).

DISCUSSION

Data presented in this work demonstrate that the utilization of nuclear-targeted Green Fluorescent Protein together with its detection via confocal microscopy allows the monitoring of the shape and behaviour of nuclei within practically all *Arabidopsis* tissues and organs. Observations are made *in vivo* without disturbance of the plant tissues by fixation and/or staining. Transgenic expression of nuclear-targeted GFP does not appear to adversely affect plant morphology, growth and development. Similar observations have been made by other workers (Pang *et al.*, 1996; Stewart, 1996; Grebenok *et al.*, 1997*a*, *b*) who reported the accumulation of very high levels of GFP fluorescence within nuclei without obvious deleterious effects. This is not consistent with the suggestion that GFP might act as a source of free radicals in the cell, and that GFP accumulation within the nucleus might result in DNA damage (Haseloff *et al.*, 1997). It should be noted that it is not possible to directly compare the levels of GFP within the various different transgenic plants produced by these groups. Furthermore, any potentially phototoxic effects of GFP expression and targeting are most likely a function of environmental variables, including illumination intensity, quality and photoperiod, which again are not directly comparable between the different groups. For these reasons, further experiments are necessary to define the extent of toxic effects accompanying GFP expression.

Constitutive expression of nuclear-targeted GFP

In plants transformed with the pBGF-D35S construct, GFP expression was observed in all cell types examined. However, the levels of expression varied substantially, being higher in younger tissues than in older ones. These findings are in agreement with those of Williamson *et al.* (1989), who found that younger, actively growing leaf, stem, root, and flower tissues of transgenic tobacco contained higher steady state levels of zein RNA expressed under the regulation of CaMV 35S promoter than did older, more quiescent tissues.

Expression of nuclear-targeted GFP regulated by the D35S promoter allowed a systematic survey of nuclear morphology in individual tissues and cell types. The shape of nuclei was mostly dependent on the location and specific function of the cell. The most commonly observed shape (a 'typical nuclear shape': Hall, Flowers and Roberts, 1974) was roughly spherical. This nuclear morphology is characteristic for meristematic and undifferentiated cells. As the cells acquired specific, non-isometric shapes during differentiation, the position of their nuclei usually changed. This was probably due, in part, to the enlargement of the vacuoles, which pushes the nucleus close to the cell wall (Hall *et al.*, 1974; Gunning and Steer, 1975). The shape of the nuclei in differentiated cells was mostly related to the shape of the cell, so if the cell was isodiametric, the nucleus was generally spherical, while in cylindrical or highly elongated cells the nuclei were ellipsoidal or rod-like. This tendency has been noted previously (De Robertis, Nowinski and Saez, 1965; Clowes and Juniper, 1968).

A substantial portion of *Arabidopsis* cells undergoes endoreduplication during normal development (Galbraith, Harkins and Knapp, 1991). The resulting differences in nuclear DNA content are reflected by their sizes, and can be readily visualized in the transgenic plants. The nuclei of guard cells, which are known to be diploid (Melaragno, Mehrothra and Coleman, 1993), are clearly the smallest, in contrast with surrounding, endoreduplicated cells whose nuclei are readily recognized as being larger. The largest nuclei, observed within trichomes, are known to undergo multiple rounds of endoreduplication (Melaragno *et al.*, 1993; Hulskamp, Misera and Jurgens, 1994; Marks, 1997). They consequently appear much larger than the rest of the nuclei within the plant, which are predominantly diploid and tetraploid (Galbraith *et al.*, 1991; Melaragno *et al.*, 1993).

Our observations confirmed that movement of nuclei in cells was active, but predominantly non-directional, presumably being driven by cytoplasmic streaming (Gunning and Steer, 1975; Williamson, 1993). Haseloff *et al.* (1997) also observed cytoplasmic streaming and organellar movement in transgenic *Arabidopsis* cells accumulating GFP within the lumen of the endoplasmic reticulum. GFP can be thus used for easy visualization of cytoplasmic streaming and any changes which might accompany alterations to the cytoskeleton (Gunning and Hardham, 1982). An exception to non-directional movement was found in the root hairs, where the movement of nuclei was both directed and developmentally-regulated. Movement of the nucleus seems to be a integral part of development of root hair cells (Williamson, 1993). Microtubules and F-actin are involved in the process of nuclear migration into the developing root

FIG. 3. Cell type-specific expression. A–C, Plants transformed using the pBGF-CoY vector. A, Fully developed primary root tip, with expression being observed within the cells of the vascular cylinder as well as the root cap. Bar = 50 μm. B, A young lateral root tip within which GFP expression is restricted to the cells of the vascular cylinder. Bar = 50 μm. C, Cotyledons and stem. Bar = 100 μm. D, Flower buds. Bar = 50 μm. E, Expression of GFP within the cells of the vascular bundle of stamens. Bar = 50 μm. F–G, Plants transformed using the pBGF-KAT vector. F, Leaf epidermis. Bar = 25 μm. G, Pistil. Bar = 25 μm. Arrows indicate expression of GFP within the cells of the style, and a lack of expression within the cells of the carpels.

hair, the actin probably moving the nucleus in a basal direction (Schnepf, 1986; Williamson, 1993). The precise mechanism of migration towards the tip has not yet been elucidated. Use of plants with nuclei highlighted by GFP accumulation should greatly assist the acquisition of information about this process. Root hair cells are often used in studies of the development of root epidermal patterning. Many mutants with defects in hair formation have been identified and studied (Dolan and Roberts, 1995; Dolan, 1996; Scheres and Wolkenfelt, 1998). Some of these mutants are affected in the processes of root hair initiation and elongation (Schiefelbein and Benfey, 1994). Crossing these mutants to our lines accumulating nuclear GFP should provide a simple means to characterize the influence of these mutations on cellular morphogenesis.

The final use of plant lines constitutively accumulating nuclear GFP involved precise monitoring of the process of cell division *in vivo*. This monitoring is facilitated by the further observation that it was easy to predict, on the basis of nuclear morphology, which cells within the root tip were at the point of division. Their nuclei were the largest within the cortical layer, and had a characteristic appearance, being well separated from adjacent nuclei. This is in concordance with the ideas that attaining a specific cytoplasm volume is a critical step for cell division (Francis and Halford, 1995), and that there is a correlation between the size of nuclei and the cytoplasmic volume (Clowes and Juniper, 1968).

Cell type-specific expression

Expression regulated by CoYMV promoter. The phloem-specific expression pattern of GFP regulated by CoYMV promoter in transgenic *Arabidopsis* was in general very similar to that described for tobacco (Medberry *et al*., 1992) and oat (Torbert *et al*., 1998). For example, expression levels were greatest in young tissues and plants, decreasing with plant and tissue age. However, there were several differences: in anthers, GFP expression was limited to the vascular bundle between the anther lobes. In comparison, for tobacco anthers, CoYMV-regulated transgenic GUS expression was observed in both non-vascular and vascular tissues (Medberry *et al*., 1992), and, for oat anthers, no GUS expression was detected (Torbert *et al*., 1998). This suggests divergence of expression patterns or of the binding properties of transcription factors interacting with CoYMV promoter in these three species.

Whereas most of the transgenic *Arabidopsis* plants displayed similar phenotypes, there were some exceptions. In one line, GFP expression was also observed in the leaf trichomes; this might be due to the integration of the transgene close to enhancers or other sequences regulating trichome-specific gene expression. In several lines, GFP expression was absent from the vascular tissues of roots, only being observed in non-vascular tissues. However, expression in all other tissues was phloem-specific. This phenomenon could be explained by truncation of the promoter sequence during insertion of T-DNA into the plant genome, thereby eliminating regulatory sequences responsible for correct promoter activity in root vascular tissues. It is known that CaMV 35S promoter is comprised from a series of modular subdomains which, when separated, confer tissue-specific expression (Benfey, Ren and Chua, 1989, 1990). One of these modules is known to direct expression principally in roots. Similarly, the promoter of the rice tungro bacilliform virus (RTBV) comprises several elements which act synergistically to confer tissue-specific gene expression (Chen *et al*., 1994, 1996; Yin, Chen and Beachy, 1997). Plant nuclear factors bind to these elements specifically and activate transcription in a tissue-specific manner (Yin *et al*., 1997). One of these transcription factors was found to be specific for shoots and deletion of sequence to which this factor is binding caused very reduced activity of the promoter in this tissue (Yin and Beachy, 1995). Moreover, there are differences in the binding of nuclear factors from shoots and from roots, which suggest different activities of the promoter in these organs (Yin and Beachy, 1995). Since RTBV is a member of the same group of viruses as CoYMV and since the RTBV genome bears similarities to that of the caulimoviruses (Qu *et al*., 1991), it seems likely that similar modular domains conferring tissue-specific expression exist within the CoYMV promoter.

Expression regulated by KAT1 promoter. Transgenic expression of GFP under the control of the KAT1 promoter was found to be specific for the guard cells. This is in contrast to the findings of Nakamura *et al*. (1995), who reported additional KAT1-regulated transgenic GUS expression in the root tissues of two transformed lines. These two lines had stronger expression than others, in which gene expression was strictly limited to the guard cells. The authors proposed that the observation of expression in roots might be a consequence of the low level activity of the KAT1 promoter in roots, coupled to elevated levels of GUS substrate transported into root cells. However, we did not observe GFP expression in roots even for lines that exhibited very high levels of guard cell expression. Our data are consistent with the results of Butt, Blatt and Ainsworth (1997), who employed PCR amplification following reverse transcription as a means to monitor expression of the native KAT1 gene; they observed expression in leaves and flowers but not in roots.

Expression conferred by the KAT1 promoter was uniform in all tissues that contained mature guard cells. In the case of the pistil, expression in this tissue was observed only in the style. In most lines, there was no expression in the remainder of the pistil; very slight expression was detected in those lines exhibiting the highest overall levels of GFP expression elsewhere in the plant. This is consistent with the established chronology of pistil development, in which carpel differentiation follows that of the style and stigma. We examined flowers of the transgenic plants for expression of GFP immediately prior to fertilization. At this time, the stomata on the style are fully developed, but there is little evidence of stomata on the carpels (Bowman, 1994); the stigma lacks stomata at all times. This probably explains why no expression was observed in carpels.

Although the patterns of expression were largely consistent, one transgenic line displayed an unusual, variable pattern of GFP expression and accumulation; in some stomata, GFP expression and nuclear accumulation was

seen in both guard cell nuclei. In other cases, it was restricted to one of the two guard cell nuclei; in yet other stomata, GFP expression was absent. This behaviour may be due to transgene silencing (Matzke and Matzke, 1995).

CONCLUSIONS

We have demonstrated the utility of the phenotype of nuclear-targeted GFP for a variety of studies in plant biology. In terms of potential further uses of the nuclear targeted GFP phenotype as an analytical tool in plant biology, a number of systems come immediately to mind. In terms of nuclear morphology, this phenotype would allow precise temporal and spatial analysis of events involved in programmed cell death (Pennell and Lamb, 1997), either occurring as part of development or as a response to biotic and abiotic environmental stimuli. In terms of the analysis of nuclear movement *in vivo*, studies of events surrounding pollination and fertilization (O'Neill, 1997; Taylor and Hepler, 1997) would greatly benefit from use of this phenotype placed under the control of appropriate tissue-specific promoters. The process whereby *Rhizobium* establishes its symbiotic relationship with legume roots (Mylona, Pawlowski and Bisseling, 1995) would also benefit from an ability to highlight the position of the legume nuclei in time and space. Finally, in terms of analysis of the regulation of cell division, the fluorescent nuclear phenotype should contribute considerably to our understanding of a number of processes including fertilization, *Rhizobium*-mediated root nodule development, *Agrobacterium* transformation, and tissue-specific endoreduplication (De Rocher *et al.*, 1990). The material for the majority of these studies can be obtained simply by crossing the plants carrying mutant phenotypes with the lines described in this paper or by just using these lines as primary experimental material. All the lines described here are available upon request.

ACKNOWLEDGEMENTS

This work was supported by a grant to D.W.G. from the U.S.D.A. N.R.I.C.G.P. (Plant Genome). We thank Georgina Lambert for valuable technical assistance.

LITERATURE CITED

Baulcombe DC, Chapman S, SantaCruz S. 1995. Jellyfish green fluorescent protein as a reporter for virus infections. *Plant Journal* **7**: 1045–1053.

Bechtold N, Ellis J, Pelletier G. 1993. *In planta Agrobacterium* mediated gene transfer by infiltration of adult *Arabidopsis thaliana* plants. *Comptes Rendus de l'Academie des Sciences. Serie III Sciences de la Vie* **316**: 1194–1199.

Benfey PN, Chua N-H. 1990. The cauliflower mosaic virus 35S promoter: Combinatorial regulation of transcription in plants. *Science* **250**: 959–966.

Benfey PN, Ren L, Chua N-H. 1989. The CaMV 35S enhancer contains at least two domains which can confer different developmental and tissue-specific expression patterns. *EMBO Journal* **8**: 2195–2202.

Benfey PN, Ren L, Chua N-H. 1990. Tissue-specific expression from CaMV 35S enhancer subdomains in early stages of plant development. *The EMBO Journal* **9**: 1677–1684.

Bent AF, Kunkel BN, Dahlbeck D, Brown KL, Schmidt R, Giraudat J, Leung J, Staskawicz BJ. 1994. RPS2 of *Arabidopsis thaliana*: a leucine-rich repeat class of plant disease resistance genes. *Science* **265**: 1856–1860.

Bevan M. 1984. Binary *Agrobacterium* vectors for plant transformation. *Nucleic Acids Research* **12**: 8711–8721.

Bowman J. 1994. Arabidopsis, *an atlas of morphology and development*. Berlin: Springer-Verlag.

Butt AD, Blatt MR, Ainsworth CC. 1997. Expression, evolution and genomic complexity of potassium ion channel genes of *Arabidopsis thaliana*. *Journal of Plant Physiology* **150**: 652–660.

Chalfie M, Tu Z, Euskirchen G, Ward WW, Prasher DC. 1994. Green fluorescent protein as a marker for gene expression. *Science* **263**: 802–805.

Chen G, Mueller M, Potrykus I, Hohn T, Futterer J. 1994. Rice tungro bacilliform virus: transcription and translation in protoplasts. *Virology* **204**: 91–100.

Chen G, Rothnie HM, He X, Hohn T, Futterer J. 1996. Efficient transcription from the rice tungro bacilliform virus promoter requires elements downstream of the transcription start site. *Journal of Virology* **70**: 8411–8421.

Chiu WL, Niwa Y, Zeng W, Hirano T, Kobayashi H, Sheen J. 1996. Engineered GFP as a vital reporter in plants. *Current Biology* **6**: 325–330.

Clowes FAL, Juniper BE. 1968. *Plant cells*. Oxford and Edinburgh: Blackwell Scientific Publications.

De Robertis EDP, Nowinski WW, Saez FA. 1965. *Cell biology*. Philadelphia and London: W. B. Saunders Company.

De Rocher EJ, Harkins KR, Galbraith DW, Bohnert HJ. 1990. Developmentally regulated systemic endopolyploidy in succulents with small genomes. *Science* **250**: 99–101.

Dolan L. 1996. Pattern in the root epidermis: An interplay of diffusible signals and cellular geometry. *Annals of Botany* **77**: 547–553.

Dolan L, Roberts K. 1995. The development of cell pattern in the root epidermis. *Philosophical Transactions of the Royal Society of London. Series B. Biological Sciences* **350**: 95–99.

Francis D, Halford NG. 1995. The plant cell cycle. *Physiologia Plantarum* **93**: 365–374.

Galbraith DW, Grebenok RJ, Lambert GM, Sheen J. 1995. Flow cytometric analysis of transgene expression in higher plants: Green Fluorescent Protein. *Methods in Cell Biology* **50**: 3–14.

Galbraith DW, Lambert GM. 1996. Advances in the flow cytometric characterization of plant cells and tissues. In: Al-Rubeai M, Emery AN, eds. *Flow cytometry applications in cell culture*. New York, Marcel Dekker, Inc., 311–326.

Galbraith DW, Harkins KR, Jefferson RA. 1988. Flow cytometric characterization of the chlorophyll contents and size distributions of plant protoplasts. *Cytometry* **9**: 75–83.

Galbraith DW, Harkins KR, Knapp S. 1991. Systemic endoploidy in *Arabidopsis thaliana*. *Plant Physiology* **96**: 985–989.

Galbraith *et al.* 1995.

Grebenok RJ, Lambert GM, Galbraith DW. 1997*a*. Characterization of the targeted nuclear accumulation of GFP within the cells of transgenic plants. *Plant Journal* **12**: 685–696.

Grebenok RJ, Pierson E, Lambert GM, Gong F-C, Afonso CL, Haldeman-Cahill R, Carrington JC, Galbraith DW. 1997*b*. Green-fluorescent protein fusions for efficient characterization of nuclear targeting. *Plant Journal* **11**: 573–586.

Gunning BES, Hardham AR. 1982. Microtubules. *Annual Review of Plant Physiology* **33**: 651–698.

Gunning BES, Steer MW. 1975. *Ultrastructure and the biology of plant cells*. London: Edward Arnold.

Hall JL, Flowers TJ, Roberts RM. 1974. *Plant cell structure and metabolism*. London: Longman.

Haseloff J. 1999. GFP variants for multispectral imaging of living cells. *Methods in Cell Biology* **58**: 139–151.

Haseloff J, Amos B. 1995. GFP in plants. *Trends in Genetics* **11**: 328–329.

Haseloff J, Siemering KR, Prasher DC, Hodge S. 1997. Removal of a cryptic intron and subcellular localization of green fluorescent protein are required to mark transgenic *Arabidopsis* plants brightly. *Proceedings of the National Academy of Sciences of the USA* **94**: 2122–2127.

Hulskamp M, Misera S, Jurgens G. 1994. Genetic dissection of trichome cell development in *Arabidopsis*. *Cell* **76**: 555–566.

Inoué S, Oldenbourg R. 1998. Microtubule dynamics in mitotic spindle displayed by polarized light microscopy. *Molecular Biology of the Cell* **9**: 1603–1607.

Knebel W, Quader H, Schnepf E. 1990. Mobile and immobile endoplasmic reticulum in onion bulb epidermis cells: short- and long-term observations with a confocal laser scanning microscope. *European Journal of Cell Biology* **52**: 328–340.

Köhler RH, Cao J, Zipfel WR, Webb WW, Hanson MR. 1997*a*. Exchange of protein molecules through connections between higher plant plastids. *Science* **276**: 2039–2042.

Köhler RH, Zipfel WR, Webb WW, Hanson MR. 1997*b*. The green fluorescent protein as a marker to visualize plant mitochondria *in vivo*. *Plant Journal* **11**: 613–621.

Leech RM, Marrison JL. 1996. Immunofluorescent quantitation of chloroplast proteins. *Plant Journal* **10**: 1169–1175.

Marks MD. 1997. Molecular genetic analysis of trichome development in *Arabidopsis*. *Annual Review of Plant Physiology and Plant Molecular Biology* **48**: 137–163.

Matzke MA, Matzke AJM. 1995. How and why do plants inactivate homologous (trans)genes? *Plant Physiology* **107**: 679–685.

Medberry SL, Lockhart BEL, Olszewski NE. 1992. The *Commelina* yellow mottle virus promoter is a strong promoter in vascular and reproductive tissues. *Plant Cell* **4**: 185–192.

Melaragno JE, Mehrotra B, Coleman AW. 1993. Relationship between endopolyploidy and cell size in epidermal tissue of *Arabidopsis*. *Plant Cell* **5**: 1661–1668.

Murashige T, Skoog F. 1962. A revised medium for rapid growth and bio assays with tobacco tissue cultures. *Physiologia Plantarum* **15**: 473–497.

Mylona P, Pawlowski K, Bisseling T. 1995. Symbiotic nitrogen fixation. *Plant Cell* **7**: 869–885.

Nakamura RL, McKendree WL, Hirsch RE, Sedbrook JC, Gaber RF, Sussman MR. 1995. Expression of an *Arabidopsis* potassium channel gene in guard cells. *Plant Physiology* **109**: 371–374.

O'Neill SD. 1997. Pollination regulation of flower development. *Annual Review of Plant Physiology and Plant Molecular Biology* **48**: 547–574.

Oparka KJ, Roberts AG, Santa Cruz S, Boevink P, Prior DAM, Smallcombe A. 1997. Using GFP to study virus invasion and spread in plant tissues. *Nature* **388**: 401–402.

Pang S-Z, DeBoer DL, Wan Y, Ye G, Layton JG, Neher MK, Armstrong CL, Fry JE, Hinchee MAW, Fromm ME. 1996. An improved green fluorescent protein gene as a vital marker in plants. *Plant Physiology* **112**: 893–900.

Pennell RI, Lamb C. 1997. Programmed cell death in plants. *Plant Cell* **9**: 1157–1168.

Pyke KA, Leech RM. 1994. A genetic analysis of chloroplast division and expansion in *Arabidopsis thaliana*. *Plant Physiology* **104**: 201–207.

Qu RD, Bhattacharyya M, Laco GS, De Kochko A, Rao BL, Kaniewska MB, Elmer JS, Rochester DE, Smith CE, Beachy RN. 1991. Characterization of the genome of rice tungro bacilliform virus: comparison with Commelina yellow mottle virus and caulimoviruses. *Virology* **185**: 354–364.

Reichel C, Mathur J, Eckes P, Langenkemper K, Koncz C, Schell J, Reiss B, Maas C. 1996. Enhanced green fluorescence by the expression of an *Aequorea victoria* green fluorescent protein mutant in mono- and dicotyledonous plant cells. *Proceedings of the National Academy of Sciences of the USA* **93**: 5888–5893.

Rouwendal GJA, Mendes O, Wolbert EJH, DeBoer AD. 1997. Enhanced expression in tobacco of the gene encoding green fluorescent protein by modification of its codon usage. *Plant Molecular Biology* **33**: 989–999.

Scheres B, Wolkenfelt H. 1998. The *Arabidopsis* root as a model to study plant development. *Plant Physiology and Biochemistry* **36**: 21–32.

Schiefelbein JW, Benfey PN. 1994. Root development in *Arabidopsis*. In: Meyerowitz EM, Somerville CR, eds. *Arabidopsis*. New York: Cold Spring Harbor Laboratory Press, 335–353.

Schnepf E. 1986. Cellular polarity. *Annual Review of Plant Physiology* **37**: 23–47.

Sheen J, Hwang S, Niwa Y, Kobayashitl, Galbraith DW. 1995. Green-fluorescent protein as a vital marker in plant cells. *Plant Journal* **8**: 777–784.

Stewart CN. 1996. Monitoring transgenic plants using *in vivo* markers. *Nature Biotechnology* **14**: 682.

Sullivan KF, Kay SA. 1999. *Green Fluorescent Proteins. Methods in cell biology, volume 58*. San Diego: Academic Press.

Taylor LP, Hepler PK. 1997. Pollen germination and tube growth. *Annual Review of Plant Physiology and Plant Molecular Biology* **48**: 461–491.

Torbert KA, Gopalraj M, Medberry SL, Olszewski NE, Somers DA. 1998. Expression of the *Commelina* yellow mottle virus promoter in transgenic oat. *Plant Cell Reports* **17**: 284–287.

Tsien RY. 1998. The Green Fluorescent Protein. *Annual Review of Biochemistry* **67**: 509–544.

Wang S, Hazelrigg T. 1994. Implications for *bcd* mRNA localization from spatial distribution of *exu* protein in *Drosophila* oogenesis. *Nature* **369**: 400–403.

Williamson RE. 1993. Organelle movements. *Annual Review of Plant Physiology and Plant Molecular Biology* **44**: 181–202.

Williamson JD, Hirsch-Wyncott ME, Larkins BA, Gelvin SB. 1989. Differential accumulation of a transcript driven by the CaMV 35S promoter in transgenic tobacco. *Plant Physiology* **90**: 1570–1576.

Wu F-S. 1987. Localization of mitochondria in plant cells by vital staining with rhodamine 123. *Planta* **171**: 346–357.

Yin Y, Beachy RN. 1995. The regulatory regions of the rice tungro bacilliform virus promoter and interacting nuclear factors in rice (*Oryza sativa* L.). *Plant Journal* **7**: 969–980.

Yin Y, Chen L, Beachy RN. 1997. Promoter elements required for phloem-specific gene expression from the RTBV promoter in rice. *Plant Journal* **12**: 1179–1188.

Yin Y, Zhu Q, Dai S, Lamb C, Beachy RN. 1997. RF2a, a bZIP transcriptional activator of the phloem-specific rice tungro bacilliform virus promoter, functions in vascular development. *EMBO Journal* **16**: 5247–5259.